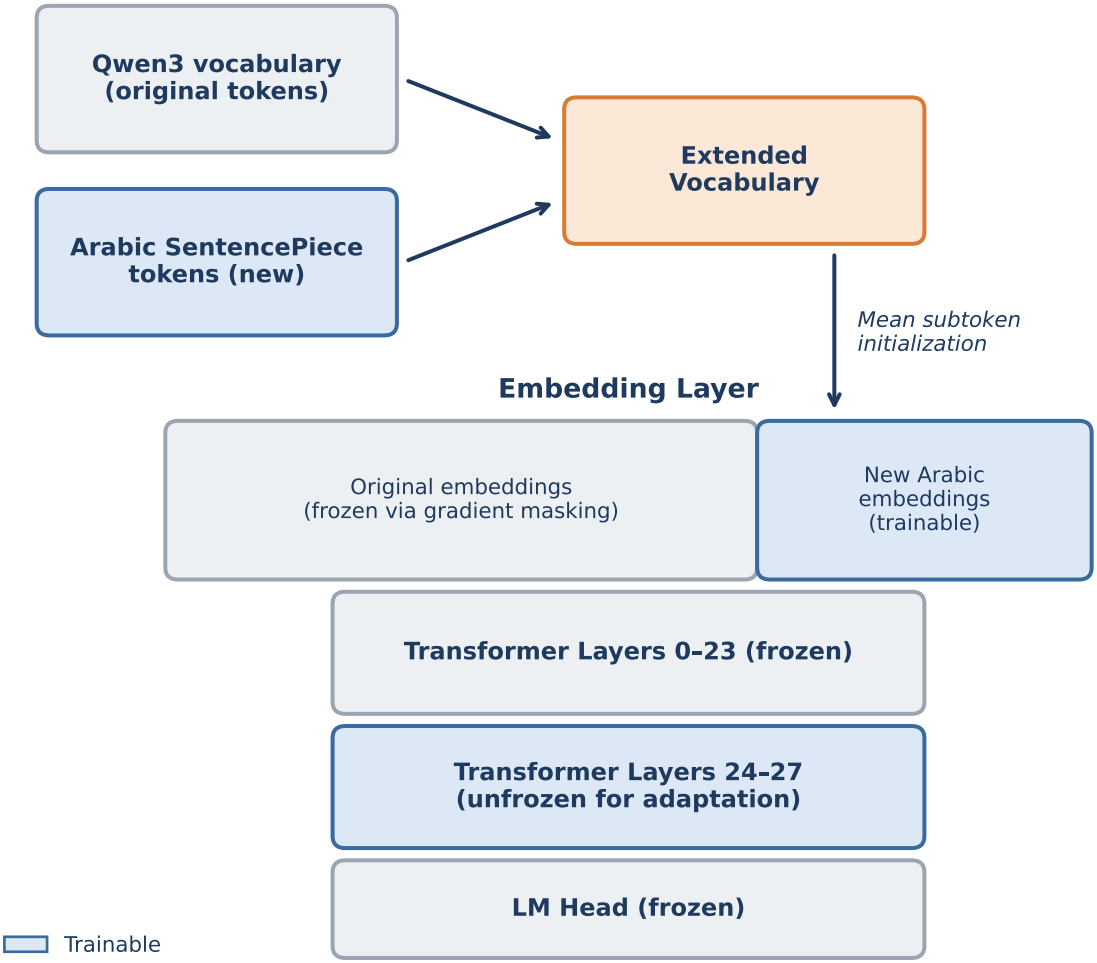
Qwen3 vocabulary
(original tokens)
Arabic SentencePiece
tokens (new)
Extended
Vocabulary
Mean subtoken
initialization
Embedding Layer
Original embeddings
(frozen via gradient masking)
New Arabic
embeddings
(trainable)
Transformer Layers 0–23 (frozen)
Transformer Layers 24–27
(unfrozen for adaptation)
LM Head (frozen)
Trainable
Frozen